

ME594: 3D Robotics, HW 8

Homework due on 03-16-2025, Topics: Dynamics. Please read each question on how to submit the work.

1. **Equations of motion of cart-pole system:** Consider a cart-pole system. The mass of the cart is m_1 . The pole has a mass m_2 at a distance of ℓ and an inertia of zero. The pole is at an angle of θ to the vertical. The linear velocity of the cart is v_x , and the angular velocity of the pole is ω . Gravity is g and points downwards. If the equations of motion are given by

$$\mathbf{M}\ddot{\Theta} + \mathbf{C} + \mathbf{G} = 0$$

where $\ddot{\Theta} = [\ddot{x}, \ddot{\theta}]$ is the acceleration, $\mathbf{M} \in \mathbb{R}^{2 \times 2}$ (this means that $\mathbf{M}$ is a matrix that belongs to the set of real numbers dimension 2 by 2) is equivalent mass matrix, $\mathbf{G} \in \mathbb{R}^2$ are all terms containing gravity, and $\mathbf{C} \in \mathbb{R}^2$ are all terms containing the velocities. Compute symbolic values of $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{G}$ using either the Euler-Lagrange method or the Recursive Newton-Euler Method.

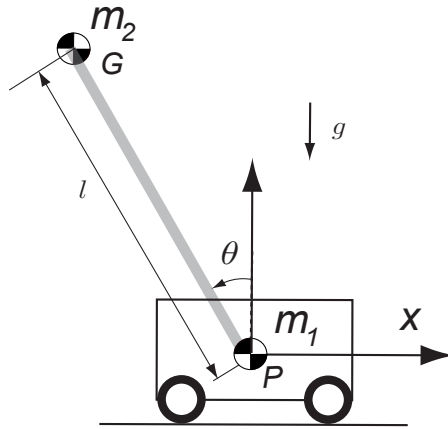


Figure 1: A cart-pendulum

- **Gradescope:** Submit your final answer and include your calculations. Hand calculations and/or symbolic computations are both acceptable.
- **Email:** None

2. **Dynamics of the Trossen Robotics WidowX 250 S:** Figure 2 shows the WidowX 250S by Trossen Robotics. A video of the robot in action is here: https://www.youtube.com/watch?v=l_10IH0JNxI. I have provided the wx250s_no_gripper.xml files that excludes the gripper. Please use this file and include it in the scene.xml meshes that can be downloaded from GitHub https://github.com/google-deepmind/mujoco_menagerie/tree/main/trossen_wx250s

Given the following position and velocity for the first six joints

$q = np.array([0.5123, -1.6532, 0.5439, -0.3432, -0.1125, 0.2342])$

$u = np.array([0.2291, -0.8602, -0.8068, -0.7110, 0.0689, 0.457029])$. If the equations of motion are given by

$$\mathbf{M}\ddot{\mathbf{\Theta}} + \mathbf{C} + \mathbf{G} = 0,$$

where $\ddot{\mathbf{\Theta}} \in \mathbb{R}^6$ is the acceleration (Here $\in \mathbb{R}^6$ means belongs to 6-dimensional real vector space), $\mathbf{M} \in \mathbb{R}^{6 \times 6}$ is the equivalent mass matrix (Here $\in \mathbb{R}^{6 \times 6}$ means belongs to 6 times 6 real-valued matrices), $\mathbf{G} \in \mathbb{R}^6$ are all terms containing gravity, and $\mathbf{C} \in \mathbb{R}^6$ are all terms containing the velocities. Compute the numerical values of $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{G}$ for the given joint angles $\mathbf{q}$ and joint velocity $\mathbf{u}$ using the Recursive Newton-Euler Method.

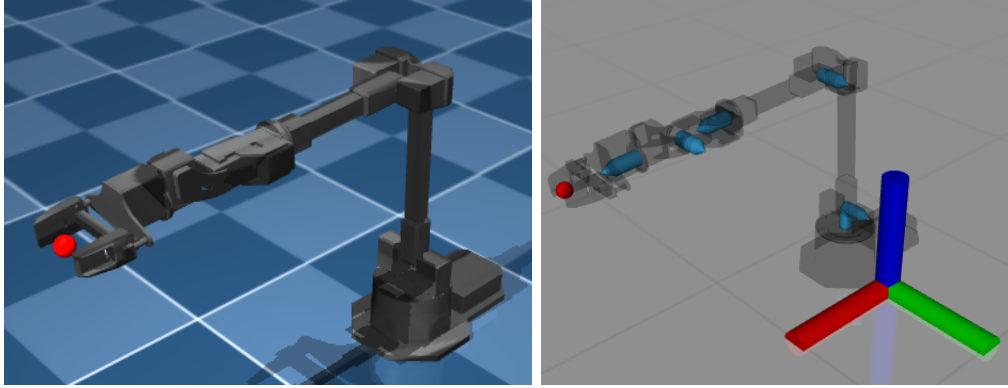


Figure 2: Perspective view of the Trossen WidowX 250 S: The robot has 6 joints, as shown in the figure on the right. The world frame is at the base of the WidowX but moved to the side in the figure to avoid clutter. The body frames are in the same direction as the world frame and centered at the joints.

Submission:

- **Gradescope:** None.
- **Email:** Submit a zipped folder: *w250s_dynamics_names.zip* where *names* should be replace by names of the team member(s) and sent to pranav@uic.edu. Please add “HW8 wx250s” to your subject line. If submitting as a group, please copy your group member.